

JORGE LEON

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EDUCATION

California Baptist University, Riverside, CA

Bachelor of Science in Electrical and Computer Engineering (Software Engineering Minor)

ABET Accredited Program

Expected: May 2027

ENGINEERING EXPERIENCE

Ontrack Technology Group, Inc.

May 2025 – August 2025

Software Engineering Intern

- Contributed to a real-time operations dispatch platform used at large-scale events with 10,000–60,000 attendees and over 20 response teams. Worked with PostgreSQL database and WebSocket communications.
- Developed front-end components for incident routing and a real-time monitoring dashboard using Angular, helping reduce average response times by 21%.
- Delivered software that reliably processed 500+ incidents per weekend with 99.7% system uptime.

CalBaptist Racing – Formula SAE EV Team

September 2023 – May 2024

Electronics Subsystem (Hardware)

- Contributed to the design and manufacturing of printed circuit boards for vehicle systems through Altium Designer, including the Electrical Distribution Panel, Driver Control module, and the high-voltage board.
- Designed 2D wiring harness diagrams and assembled the electrical system across the vehicle, routing wires, sensors, and connectors in strict compliance with FSAE rules.
- Coordinated with cross-functional teammates (different subsystems) using shared Microsoft Excel workbooks for expense tracking, parts inventory management, and work order status updates.

PROJECTS

Arduino Uno Rev3 PCB Redesign

January 2026 - Present

- Designed a complete two-layer custom PCB layout for the Arduino Uno Rev3 in Altium Designer, including schematic capture, component placement, multi-layer routing, and design rule validation with zero errors.
- Generated production-ready Gerber files and assembly documentation for board fabrication through JLCPCB.
- Performed electromagnetic and signal integrity analysis using ANSYS SIwave including density plots, crosstalk analysis, and trace impedance evaluation to identify layout improvements.

Response Unit Tracker (GPS-Based)

June 2025 – September 2025

- Modified embedded firmware to build a complete GPS-based tracking system for response teams in the field.
- Deployed scalable mesh network infrastructure ranging from 4 to 20 nodes serving venues up to 60,000 patrons, resulting in a 28% improvement in response times through real-time location tracking.
- Integrated microcontrollers, wireless transceivers, and GPS modules in areas with limited or no cellular coverage.

WORK EXPERIENCE

FedEx Freight – Logistics Analyst

February 2025 - Present

Analyze transportation data and logistics metrics using Microsoft Excel and in-house database systems to optimize freight routing efficiency and identify cost-reduction opportunities across the distribution center.

Prevent Medical Solutions – Command Center Operations

October 2022 – Present

Served as a primary point of contact for event organizers, venue security, and medical teams, relaying critical updates and coordinating multi-agency response when needed.